PADEE.AI

UI/UX Specification

Phase 1 Build Guide -- Version 3.0

Document type: UI/UX Prototype Specification | Version 3.0 | April 2026 | Audience: Developer (founder) + Figma designer

Screens	Changed from v2	New in v3	Unchanged from v2
19 screens total	5 screens updated	2 new screens	12 screens unchanged

What Changed from v2 to v3

Five screens are updated. Two screens are new. All other v2 screens are unchanged and remain valid. This document specifies only the changes and additions -- refer to Padee_UI_Spec_v2 for unchanged screens.

CHANGED IN v3

Screen 03 -- Onboarding subject selection

v2: Only CBSE Class 9-10, Science + Math

v3: CBSE Class 8-12, all subjects. Class 11-12 adds Commerce subjects. Track selector added (School / Competitive / CA).

CHANGED IN v3

Screen 06 -- Ask AI active chat

v2: Single-step AI response, no quality signals

v3: Dual-loop thinking state (2 phases), Visual Explain chip, thumbs up/down quality signal, Report incorrect button, daily usage cap counter, NCERT source citation, memory-aware indicator.

CHANGED IN v3

Screen 15 -- Worksheet generator

v2: Custom generation only, no paper upload

v3: Two modes: Custom and Paper Mimic. PDF upload for CBSE board paper. Validation state (NCERT checking). Per-question validation badge.

NEW IN v3

Screen 19 -- Visual Explanation overlay

New screen/modal triggered by Explain visually chip. AI-generated HTML concept visualisation renders inline in a sandboxed panel within the Ask AI view.

NEW IN v3

Screen 20 -- Admin: NCERT content upload

Password-protected admin panel at padee.ai/admin. Founder-only. Upload NCERT PDFs by subject and class. Tracks ingestion progress, chunk count, and embedding status.

Unchanged screens -- refer to v2 spec

Screen	Name
01	Splash / launch screen
02	Onboarding welcome (two-column desktop layout, animated demo loop)
04	Student home -- all 10 AI-curated modules
05	Ask AI cold start (orb, 6-action grid, recent doubts, photo upload)
07	Learn -- chapter view (subject pills, chapter accordion, topic actions)

08	Practice -- MCQ (question card, 4 options, feedback slide-up)
09	Practice -- descriptive question (text area, AI evaluation card)
10	Test mode active (full-screen, timer, question nav drawer)
11	Test results (score ring, question review, AI insights, CTA strip)
12	Progress / Me tab (level bar, streak calendar, badges, weekly AI summary)
13	Gamification celebrations (level-up overlay, badge unlock bottom sheet)
14	Teacher command centre (AI alert feed, quick actions, class health, spotlight)
16	Test generator (with answer key toggle, scheduling, section structure)
17	Live class assistant (real-time MCQ gen, concept explain, lesson summary)
18	Teacher student performance view (class overview, individual AI profile)

SCREEN 03

Onboarding -- Class + Subject + Track

Updated: Class range now 8-12. Class 11-12 triggers Commerce subject options. New step 2B adds a track selector.

Step 1 of 3 -- Class selection (unchanged)

* Horizontal pill row: Class 8 / 9 / 10 / 11 / 12
* Selecting 11 or 12 triggers the Commerce subjects to appear alongside Science

Step 2 of 3 -- Subject selection (UPDATED)

Subject options now depend on class selected. Render two groups for Class 11-12.

Class	Science subjects	Commerce subjects (Class 11-12 only)
Class 8-10	Physics / Chemistry / Biology / Maths / Computer Science / English / Social Science	-- (not shown)
Class 11	Physics / Chemistry / Biology / Maths / Computer Science / English	Economics / Business Studies
Class 12	Physics / Chemistry / Biology / Maths / Computer Science / English	Economics / Accounts / Business Studies

UI behaviour: For Class 11-12, show two chip groups with headers 'Science subjects' and 'Commerce subjects'. Student can select from both groups. Subject chips use standard color coding. Commerce subject colors: Economics (amber), Accounts (blue), Business Studies (teal).

Step 3 of 3 -- Track selector (NEW)

New third step added for Phase 1. Determines home screen AI recommendations and practice difficulty.

```
What is your main goal?

[ School learning ]     I want to do well in CBSE boards
[ Foundation prep ]     I am preparing for JEE / NEET early
[ CA Foundation ]       I am planning to pursue CA

You can change this anytime from your profile.

[Continue ->]
```

* Track selection is optional -- 'Continue without selecting' skips it, defaults to School
* Foundation prep and CA Foundation unlock slightly different home screen recommendations but use the same CBSE content in Phase 1 -- advanced tracks ship in Phase 2-3
* Selected track stored in Supabase profiles.active_track column

SCREEN 06

Ask AI -- Active Chat

This is the most significantly updated screen in v3. Five additions: dual-loop thinking state, Visual Explain chip, quality signal buttons, usage cap counter, and NCERT source citation.

6.1 -- Dual-loop thinking state (NEW)

The AI runs two loops before streaming the answer. The thinking state must reflect both phases so the student understands why it takes slightly longer than a single API call.

Phase	UI display
Analysis loop	Small AI orb animates + text: 'Analysing your question...' -> 'Identifying the concept...' -> 'Checking what usually confuses students here...'
Solve loop	Text transitions to: 'Building your explanation...' -> 'Adding step-by-step breakdown...' -> response begins streaming

Total thinking time: 2-4 seconds. Do not show both loop labels simultaneously -- they transition sequentially. If the analysis completes in under 1 second, skip directly to solve loop animation without showing analysis text.

6.2 -- Action chips (UPDATED -- new chips added)

Original chips from v2 are unchanged. Two new chips are added after every AI response.

Chip	Status	Behaviour
Explain Simpler	v2 -- unchanged	AI rephrases with everyday analogy
Show Exam Answer	v2 -- unchanged	CBSE model answer with keyword highlights
Quiz Me	v2 -- unchanged	Inline MCQ with XP preview
Similar Question	v2 -- unchanged	AI generates a variation of the same problem
Challenge Me	v2 -- unchanged	Harder version of the question
Real-life Example	v2 -- unchanged	Relatable everyday analogy
Explain Visually	NEW in v3	Triggers visual explanation overlay (Screen 19). AI generates interactive HTML concept diagram.
Common Mistakes	v2 -- unchanged	What students usually get wrong here

Chip order on desktop: show all chips in a horizontal scroll row. On mobile: show first 4 chips with 'More' overflow. Always show Explain Visually within the first 4 -- it is the highest-value new chip.

6.3 -- Quality signal buttons (NEW)

Every AI response must have thumbs up / thumbs down signal buttons and a Report incorrect link. These feed the quality control pipeline.

```
[AI response bubble -- end of streaming]

[Thumb-up icon]  [Thumb-down icon]  |  Report incorrect

Placement: bottom-right of the AI bubble, below action chips
Font size: 12px, muted color, becomes active color on tap
```

* Thumb-up tap: silent, records helpful=true in doubt_sessions, no visual confirmation needed
* Thumb-down tap: shows a small inline prompt 'What was wrong? [Unclear] [Inaccurate] [Not NCERT] [Skip]' -- one tap, no modal
* Report incorrect: opens a bottom sheet with 'What is wrong with this answer?' text area + Submit -- goes to teacher review queue
* Both buttons are de-emphasised (gray/muted) until hovered or tapped. Do not make them prominent -- they should not distract from the learning content.

6.4 -- Daily usage cap indicator (NEW)

Free tier students have a 10-doubt-per-day cap. The counter must be visible but not anxiety-inducing.

```
Input bar (bottom of screen):

[Camera icon]  [  Type your doubt...          ]  [Ask AI ->]
                                          7 / 10 today

The counter '7 / 10 today' sits inside the input bar, right-aligned,
in muted gray. Changes to amber at 8/10. Changes to red with
'Last doubt today' at 10/10.

At limit (10/10):
Input bar shows: 'You have used your 10 free doubts today.'
[Upgrade to unlimited ->]  [Come back tomorrow]
```

* Premium users: no counter shown -- input bar is unchanged
* Counter resets at midnight IST -- not on rolling 24-hour basis
* Counter is only shown when the student has used 5+ doubts -- never show 1/10, 2/10 (creates anxiety early in the session)

6.5 -- NCERT source citation (NEW)

Every AI response includes a subtle citation showing which NCERT chapter the answer came from. This builds trust and proves the answer is textbook-grounded.

```
[AI response bubble -- end of content]

Source: NCERT Class 10 Physics, Chapter 12 -- Electricity

Style: 12px, muted teal color, non-tappable in Phase 1
Placement: below the main response text, above action chips
Only show when RAG retrieval returned a confident match (score > 0.85)
If no confident match: omit the citation line entirely
```

6.6 -- Memory-aware personalisation indicator (NEW)

When the AI uses the student's stored profile to personalise the response, show a small memory indicator. This makes the personalisation visible and valued rather than invisible.

```
Top of AI bubble (when memory was used):

[small orb icon]  'Remembering your profile'  (muted, 11px)

Show ONLY when: student has asked about this topic before,
OR the response references the student's weak areas or recent mistakes.
Do NOT show on every response -- only when personalisation is meaningfully used.
```

SCREEN 15

Worksheet Generator

Major addition: CBSE Paper Mimic mode. Teacher can now upload a previous year's board exam PDF and generate new questions in the same style. Also adds per-question validation badges from the worksheet validation agent.

15.1 -- Mode selector (NEW)

```
Top of the left config panel:

Mode:  [ Custom ]  [ Paper Mimic ]

'Custom' is selected by default (existing behaviour unchanged).
'Paper Mimic' switches the config panel to upload-based mode.
```

15.2 -- Paper Mimic mode -- config panel

UI element	Spec
Mode label	'Paper Mimic' pill highlighted in teal. Brief description: 'Upload last year's CBSE board paper -- I will generate new questions in the same style and mark distribution.'
PDF upload area	Large dashed upload zone: 'Drop CBSE board exam PDF here or click to upload'. Accepts PDF only. Max 20MB.
After upload	File name + size shown. Small parsing animation: 'Reading exam format...' -> 'Extracting question styles...' -> 'Analysing mark distribution...'
Class / Subject	Auto-detected from PDF when possible. Editable dropdown if detection fails.
Question count	Slider: 5 to 30 questions. Default: matches the original paper question count.
Generate button	'Generate similar questions ->' (same style as Custom mode). Text changes to 'Generating...' during processing.

15.3 -- Loading state (updated for validation)

The worksheet generation now runs a validation pass after generation. The loading state must communicate this additional step.

```
Step 1 (3-5 seconds):  AI orb + 'Creating your worksheet...'
Step 2 (2-4 seconds):  Checkmark animation + 'Checking each question against NCERT...'
Step 3 (1-2 seconds):  'Finalising...'
-> Preview loads

If any questions fail validation and are regenerated:
Show: 'Improving 2 questions...' before Step 3
```

15.4 -- Question preview (updated -- validation badges)

Each question in the preview shows a validation badge confirming NCERT accuracy.

```
Each question card in the preview panel:

Q3. (3 marks)                            [NCERT verified v]
What is the relationship between current
and voltage in a conductor?

[Edit]  [Generate similar]  [Remove]

Badge 'NCERT verified v' = small green pill, 11px, right-aligned
This badge only appears after the validation agent has passed the question.
Questions regenerated by the validation loop still show the badge after passing.
```

* Teacher can still tap [Edit] to manually modify any question -- removes the validation badge
* Teacher can tap [Report issue] on any question to flag it to the admin review queue

SCREEN 19

Visual Explanation Overlay

Triggered by the 'Explain Visually' chip in Screen 06. The AI generates a self-contained HTML visualisation of the concept. Renders inside a sandboxed panel that slides up or appears inline in the chat.

19.1 -- Trigger and loading

```
Student taps [Explain Visually] chip
-> Input bar shows: 'Generating visual explanation...'
-> AI orb pulses
-> New message type appears in chat: Visual Explanation bubble
-> Iframe renders the HTML inside the bubble
-> Total time: 3-6 seconds
```

19.2 -- Visual bubble UI

Element	Spec
Bubble header	'Visual Explanation' label + small AI orb icon. Different visual treatment from text bubbles -- uses a teal-accented border.
Content area	sandboxed iframe, max 480px wide, height auto-sizes to content. No scrollbars. White background. Renders HTML + inline CSS + inline SVG only.
Expand button	Top-right corner of bubble: expand icon -> opens the same HTML in a centered full-screen modal with close button.
Caption	Below the iframe: short AI-generated caption text explaining what the visualisation shows.
Action chips	'Quiz Me on this' | 'Explain in words' | 'See CBSE exam answer'

19.3 -- What the AI generates

The LLM (GPT-4o-mini via LiteLLM) receives a prompt: 'Generate a self-contained HTML snippet, no external dependencies, max 480px wide, inline CSS only. Visualise [concept] for a CBSE Class [n] student. Use simple SVG shapes, colour, labels, and if possible an interactive element.'

Example concepts and their expected outputs:

Concept	Expected visual
Ohm's Law (V = IR)	SVG circuit diagram: battery, resistor, ammeter, voltmeter with labelled values. Slider for resistance shows how current changes.
Photosynthesis	Leaf cross-section: sunlight arrows in, CO_2 in, O_2 out, glucose stored. Arrow animation on hover.
Chemical equation balancing	Balance scale SVG showing reactants on left, products on right. Scales visually balance when equation is correct.
Cell division (Mitosis)	Animated cell splitting: prophase -> metaphase -> anaphase -> telophase using CSS keyframes.

Pythagoras theorem	Right triangle with coloured squares on each side. a^2 + b^2 = c^2 shown geometrically. Sliders to change a and b.

Failure state: if the HTML renders blank or with a JavaScript error, show: 'Visual explanation could not be generated for this concept. Try asking for a step-by-step explanation instead.' with fallback action chips.

SCREEN 20

Admin -- NCERT Content Upload

Password-protected admin panel at padee.ai/admin. Founder-only access. Used to upload NCERT PDFs for RAG ingestion. Not visible to students or teachers.

20.1 -- Authentication

```
URL: padee.ai/admin
Auth: Simple password check (not Supabase Auth -- separate env var: ADMIN_PASSWORD)
Session: 8-hour cookie. No MFA required for Phase 1.
If wrong password: 'Incorrect admin password' -- no lockout for Phase 1.
```

20.2 -- Main layout

Two-column layout: upload panel on the left, content library on the right.

20.3 -- Upload panel (left)

Field	Spec
Subject	Dropdown: Physics / Chemistry / Mathematics / Biology / Computer Science / English / Social Science / Economics / Accounts / Business Studies
Class	Dropdown: Class 8 / 9 / 10 / 11 / 12
Chapter number	Number input (optional). If blank, system marks chunk as full-book.
Chapter name	Text input (optional). Helps the citation display in student doubts.
PDF file	File picker. Accepts PDF only. Max 50MB per file.
Upload + Process button	'Upload and Process PDF'. Disabled until all required fields are filled.

20.4 -- Processing state

```
Click 'Upload and Process PDF'
-> Progress bar: 'Uploading PDF... (12%)'
-> 'Extracting text...'
-> 'Splitting into chunks...'  (shows chunk count: 'Found 127 passages')
-> 'Generating embeddings... (45 / 127)'  (shows progress fraction)
-> 'Storing in database...'
-> [green checkmark] 'Done! 127 passages indexed.'

If an error occurs at any step: red error message with retry button.
Background processing: page can be closed after upload starts.
Processing continues server-side.
```

20.5 -- Content library (right)

Table showing all indexed content. Columns: Subject, Class, Chapter, Chunks, Indexed date, Actions.

```
Subject       Class  Chapter           Chunks  Indexed        Actions
Physics       10     Ch.12 Electricity   127   12 Apr 2026   [Re-index] [Delete]
Mathematics   10     Full book           342   11 Apr 2026   [Re-index] [Delete]
Chemistry     10     Ch.1 Chemical Rxn    89   11 Apr 2026   [Re-index] [Delete]
```

```
Re-index: re-uploads and re-processes the same PDF (for updates)
Delete: removes all chunks for that subject/class/chapter combo

Summary row at top:
'Total indexed: 48 PDFs  |  2,847 passages  |  Last updated: Today'
```

Updated Component Library -- New in v3

All components from v2 remain valid. These five components are new.

Component	Description	Variants
Quality Signal Buttons	Thumbs up / down + Report incorrect link. Appears at bottom-right of every AI response bubble, below action chips.	Default (muted gray) / Voted-up (teal) / Voted-down (amber) / Reported (red)
Usage Cap Counter	'X / 10 today' shown inside input bar right-aligned. Only shows at 5+/10 usage.	Normal (gray) / Warning (amber, 8-9/10) / Limit (red, 10/10 blocked)
NCERT Citation Chip	Small muted teal text below AI response: 'Source: NCERT Class [n] [Subject], Chapter [n]'	Visible (confident match) / Hidden (no confident match)
NCERT Verified Badge	Small green pill on worksheet questions confirming validation agent approved this question.	Verified (green) / Removed after manual edit
Visual Explanation Bubble	Special AI message bubble with sandboxed iframe containing AI-generated HTML concept visual.	Loading / Rendered / Error (fallback text)
Memory Indicator	11px label 'Remembering your profile' at top of AI bubble when personalisation was applied.	Visible (memory used) / Hidden (generic response)

New Interaction Patterns -- v3

Pattern 09 -- Visual Explanation flow

1. Student taps 'Explain Visually' chip on an AI response.
2. Input bar shows loading state: AI orb + 'Generating visual...'
3. New message type (visual bubble) appears in the chat stream, below the original text response.
4. Iframe renders the HTML. If height < 150px (probably an error), show fallback text instead.
5. Student can tap expand icon to open in full-screen modal.
6. Action chips below visual bubble: 'Quiz Me on this' / 'Explain in words' / 'See CBSE exam answer'.

Pattern 10 -- Quality signal recording

7. Thumb up: records helpful=true. No visual confirmation. Button changes to filled/active state.
8. Thumb down: shows inline 3-option prompt: [Unclear] [Inaccurate] [Not NCERT]. One tap, no modal, auto-dismisses in 3s.
9. Report incorrect: opens bottom sheet with text area. Teacher receives this in the /admin/flagged queue.
10. Only one of thumb-up or thumb-down can be active per response. Tapping the other deselects the first.

Pattern 11 -- Usage cap reached

11. Student types a doubt and taps Ask AI when at limit (10/10).
12. Input is blocked. Input bar shows: 'You have used your 10 free doubts today.'
13. Two CTAs shown below input: 'Upgrade to unlimited (Rs.99/month) ->' and 'Come back tomorrow'.
14. 'Come back tomorrow' dismisses the cap message. Input bar re-enables at midnight IST.
15. Cap only applies to doubt solver -- practice, tests, and worksheet generation are NOT capped.

Pattern 12 -- Paper Mimic upload flow

16. Teacher selects 'Paper Mimic' mode tab in worksheet generator.
17. Drags and drops a CBSE board exam PDF (or clicks to open file picker).
18. System shows: 'Reading exam format... Extracting question styles... Analysing mark distribution...' (5-8 seconds).
19. Subject and class auto-fill from PDF metadata. Teacher confirms or corrects.
20. Teacher clicks 'Generate similar questions ->'
21. Loading state shows validation steps: 'Creating questions... Checking against NCERT... Finalising...'
22. Preview shows the generated worksheet with NCERT verified badges per question.

MVP Screen List -- Phase 1 Build Order

Updated from v2. Visual Explanation and Admin upload added. Paper Mimic added to worksheet screen.

Priority	Screen	Why it is in Phase 1 MVP
P0	03 -- Onboarding (updated)	Without class/subject selection the AI cannot be NCERT-accurate. This is the prerequisite for everything.
P0	06 -- Ask AI active (updated)	This IS the product. Dual-loop + quality signals + usage cap + citation must all be present at launch.
P0	20 -- Admin NCERT upload (new)	Without NCERT content uploaded to pgvector, RAG cannot work and all AI answers are ungrounded.
P0	04 -- Student home	The daily re-engagement surface. All 10 modules must work.
P0	15 -- Worksheet generator (updated)	Teacher adoption hinges on this. CBSE Paper Mimic is the key demo feature. Must ship with validation badges.
P0	02 -- Onboarding welcome	First impression. Animated demo loop establishes product value.
P0	14 -- Teacher command centre	Without this, teachers have no reason to return daily.
P1	19 -- Visual Explanation (new)	The WOW moment for school demos. High trust signal. Build in Week 3 after core doubt flow is stable.
P1	08 -- Practice MCQ	Second most-used student feature after doubt solving.
P1	10 -- Test mode	Needed for teacher-assigned tests. Without this, teacher value is incomplete.
P1	11 -- Test results	The payoff of the test experience. AI insights must show.
P1	05 -- Ask AI cold start	The entry point to the ask flow. Quick to build, high impact on first experience.

P2	12 -- Progress / Me tab	Important for retention but not needed in Week 1 validation.
P2	07 -- Learn chapter view	Useful for structured browsing but students primarily access via home screen.
P2	01 -- Splash screen	2 hours to build. Nice polish, not critical path.